

Sample estate: RECON_TEST on Oracle XE 11g (11.2.0.2)

Schema **RECON_TEST** · Oracle 11.2.0.2.0 · extracted 2026-09-02 15:53:05
Prepared by DevCrafter · 2026-09-03 · engine: pgrecon

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MIGRATION GRADE

Executive summary

This assessment evaluates the PL/SQL codebase supporting the client's Oracle 11.2.0.2.0 environment in preparation for a migration to PostgreSQL. The scope covers 36 database objects comprising 156 lines of PL/SQL code, from which 50 discrete findings were identified across a range of migration-relevant patterns. The codebase receives an overall grade of B, indicating a real, working project with identified hot spots rather than a trivial or a highly complex system. This places the effort within a manageable but non-trivial range, requiring focused remediation in specific areas rather than a wholesale rewrite.

The findings span 35 distinct rule categories, touching on database links, package-level state, file and large object handling, autonomous transactions, dynamic SQL, and Oracle-specific constructs such as compound triggers and BFILE columns. The estimated effort to complete the migration ranges from 91.9 to 155.5 person-days, with an expected case of 113.1 person-days. This range reflects the inherent uncertainty in remediation complexity for the higher-severity items and should be used for planning and resourcing purposes rather than as a fixed commitment.

50 FINDINGS	8 HIGH	18 MEDIUM	113 EXPECTED PERSON-DAYS
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Effort range: **92–156** person-days (4.4–7.4 person-months), expected 113 — including testing and stabilization. Assumptions are listed with the estimate; this is a scoping instrument to be calibrated per engagement, not a quote.

Estate overview

36 OBJECTS	156 PL/SQL LINES	5/5 DDL PARSED	10 UNITS DEEP-PARSED
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OBJECT TYPE	COUNT
TABLE	10
INDEX	8
TRIGGER	3
SEQUENCE	3
VIEW	2
PROCEDURE	2
TYPE BODY	1
TYPE	1
SYNONYM	1
PACKAGE BODY	1
PACKAGE	1
MATERIALIZED VIEW	1
JOB	1
FUNCTION	1

License exposure

What the database reports about its own edition, hardware, and use of separately licensed features. These are the source system's facts; contract interpretation and pricing belong in the engagement, next to the actual agreements.

Express EDITION	2 CPU CORES	1 CPU_COUNT IN FORCE
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No separately licensed option shows usage in the database's own statistics: the estate runs on the edition's base feature set.

Migration hot spots

The ten objects concentrating the most remediation weight; these decide the project's shape and deserve design attention first.

OBJECT	FINDINGS	WEIGHT	DRIVEN BY
RECON_TEST.PKG_LEDGER	6	12.3	Package-level state (5), Package initialization block (2), PL/SQL collection type (2), +3 more
RECON_TEST.ARCHIVE_NOTES	9	8.3	UTL_FILE usage (3), ROWID in code (1.5), DBMS_LOB usage (1.5), +6 more
RECON_TEST.TRG_EMP_AUDIT	3	6.0	Autonomous transaction (4), Transaction control inside PL/SQL (1.5), Sequence-assigned key via trigger (0.5)
RECON_TEST.PKG_LEDGER (body)	1	5.0	Package-level state (5)
RECON_TEST.TRG_EMP_SAL_GUARD	2	4.3	Compound trigger (4), SYSDATE (0.3)
RECON_TEST.LEGACY_NOTES.BODY	1	4.0	LONG or LONG RAW column (4)
RECON_TEST.LOOPBACK	1	3.0	Database link (3)
RECON_TEST.LEGACY_REFS.SCAN_DOC	1	3.0	BFILE column (3)
RECON_TEST.PRODUCT_SPECS.SPEC	1	3.0	XMLTYPE column (3)
RECON_TEST.T_MONEY	1	2.5	User-defined object type (2.5)

Findings and remedies

HIGH

Database link R-0BJ-01 • extension: postgres_fdw

Cross-database access needs postgres_fdw or oracle_fdw on the PostgreSQL side, plus credential management. Every statement using the link must be found and repointed.

RECON_TEST.LOOPBACKto //localhost:1521/XE as RECON_TEST

HIGH Package-level state R-PKG-01

PostgreSQL has no package variables. Scalars map to session settings via `set_config` and `current_setting` (text only); composites and collections need per-session temporary tables. Every reader and writer of the state changes.

RECON_TEST.PKG_LEDGER	2 package-level declaration(s), first at line 2
RECON_TEST.PKG_LEDGER (body)	1 package-level declaration(s), first at line 2

HIGH UTL_FILE usage R-SYS-01 • extension: orafce

PostgreSQL functions have no supported server-side file API. Move file handling to the application, or accept the operational cost of an untrusted language such as `plpython`.

RECON_TEST.ARCHIVE_NOTES	UTL_FILE (first at line 16)
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HIGH Compound trigger R-TRG-01

PostgreSQL has no compound triggers. Split the timing sections into separate trigger functions and move statement-level state into a transition table or a session variable.

RECON_TEST.TRG_EMP_SAL_GUARD	on EMP
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HIGH Autonomous transaction R-TRG-02

PostgreSQL functions cannot commit independently of their caller. Rework audit-style autonomous transactions with a `dblink` loopback, a message table drained by a worker, or `pg_background`.

RECON_TEST.TRG_EMP_AUDIT	PRAGMA AUTONOMOUS_TRANSACTION (first at line 5)
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HIGH LONG or LONG RAW column R-TYPE-01

LONG has been deprecated since Oracle 8. Map LONG to text and LONG RAW to bytea; any code reading the column piecewise must be rewritten, and only one LONG per table exists to find.

RECON_TEST.LEGACY_NOTES.BODY	LONG
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HIGH **BFILE column** R-TYPE-07

BFILE points at a file outside the database and PostgreSQL has no counterpart. Store a path or object key plus the file in application storage, and move the DIRECTORY grants and BFILENAME plumbing into the application.

RECON_TEST.LEGACY_REFS.SCAN_DOC

BFILE

MEDIUM **Scheduler or legacy job** R-OBJ-02 • extension: pg_cron

DBMS_SCHEDULER and DBMS_JOB have no engine equivalent in PostgreSQL. Recreate schedules in pg_cron and port each job action; calendar syntax must be translated by hand.

RECON_TEST.NIGHTLY_ROLLUP

scheduler job

MEDIUM **Materialized view** R-OBJ-03

PostgreSQL materialized views refresh only in full and have no refresh-on-commit. Fast-refresh views need a trigger-based design or a scheduled REFRESH with the staleness accepted.

RECON_TEST.MV_DEPT_SALARIES

materialized view

MEDIUM **User-defined object type** R-OBJ-05

Composite types carry the attributes, but member functions do not exist in PostgreSQL: each method becomes a standalone function taking the composite as its first argument, and every call site changes.

RECON_TEST.T_MONEY

object type

MEDIUM **Package initialization block** R-PKG-02

Oracle runs the block once per session on first reference, which PostgreSQL cannot replicate. Extract it into an init function guarded by a session flag and call it lazily from each entry point. Detection is heuristic; confirm by eye.

RECON_TEST.PKG_LEDGER

BEGIN blocks exceed subprograms by 1 (possible initialization block)

MEDIUM Dynamic SQL R-SRC-01

EXECUTE IMMEDIATE maps to PL/pgSQL EXECUTE, but bind syntax and error handling differ. Each dynamic statement needs a manual port and a test.

RECON_TEST.PKG_LEDGER

EXECUTE IMMEDIATE (first at line 27)

MEDIUM GOTO statement R-SRC-02

PL/pgSQL has no GOTO. The control flow must be restructured with loops, exceptions, or early returns.

RECON_TEST.FIND_MANAGER

GOTO (first at line 14)

MEDIUM BULK COLLECT R-SRC-03

PL/pgSQL has no BULK COLLECT; array_agg into an array or a plain set-based statement usually replaces the whole loop.

RECON_TEST.PKG_LEDGER

BULK COLLECT (first at line 24)

MEDIUM CONNECT BY hierarchical query R-SRC-04

Rewrite as a recursive CTE (WITH RECURSIVE). LEVEL, SYS_CONNECT_BY_PATH, and ORDER SIBLINGS BY all need manual equivalents.

RECON_TEST.V_ORG_CHART

CONNECT BY in view definition

MEDIUM Transaction control inside PL/SQL R-SRC-14

PostgreSQL functions cannot COMMIT or ROLLBACK; only procedures called outside a transaction block can. Each routine that controls transactions must become a procedure or hand that control back to the caller.

RECON_TEST.TRG_EMP_AUDIT

COMMIT or ROLLBACK (first at line 15)

MEDIUM PL/SQL collection type R-SRC-16

Nested tables, varrays, and associative arrays map to PostgreSQL arrays, jsonb, or temporary tables depending on use; collection methods such as EXTEND, COUNT, and FIRST have no direct equivalents and every reader changes.

RECON_TEST.PKG_LEDGER

collection type declaration (first at line 20)

MEDIUM Empty string treated as NULL R-SRC-18

Oracle treats "" as NULL; PostgreSQL treats it as a real value. `v := ''` assigns NULL in Oracle but an empty string in PostgreSQL, `NVL(x, '')` stops collapsing, and `col = ''` can start matching rows. Nothing errors: behavior changes silently. Decide per site whether NULL or empty was meant, and put the data path under test.

RECON_TEST.ARCHIVE_NOTES

empty-string literal (first at line 4)

MEDIUM ROWID in code R-SRC-19

ctid is not a ROWID: it moves on update and vacuum and must not be stored or compared. Fetch-by-rowid loops rewrite to primary-key access or WHERE CURRENT OF; ROWID-typed variables and parameters change type with them.

RECON_TEST.ARCHIVE_NOTES

ROWID (first at line 3)

MEDIUM DBMS_LOB usage R-SYS-04

Piecewise LOB calls mostly disappear: text and bytea are handled whole, with substring and length built in. Streaming access beyond 1 GB needs large objects and the `lo_ API`.

RECON_TEST.ARCHIVE_NOTES

DBMS_LOB (first at line 14)

MEDIUM Global temporary table R-TAB-01 • extension: pgtt

PostgreSQL temporary tables are per session and vanish with it; there is no shared definition with private rows. Rework to on-the-fly temp tables or use the pgtt extension.

RECON_TEST.STAGING_ROWS

GLOBAL TEMPORARY

MEDIUM Sequence-assigned key via trigger R-TRG-03

The pre-12c idiom of filling a key from a sequence in a BEFORE INSERT trigger becomes a plain identity column or a DEFAULT nextval() in PostgreSQL; the trigger disappears.

RECON_TEST.TRG_EMP_AUDIT	NEXTVAL in trigger (first at line 14)
RECON_TEST.TRG_INVOICES_ID	NEXTVAL in trigger (first at line 6)

MEDIUM XMLTYPE column R-TYPE-02

PostgreSQL xml stores documents but has no XML index type and none of the XMLType member functions; XQuery support differs. Inventory how the application queries the XML before mapping.

RECON_TEST.PRODUCT_SPECS.SPEC	XMLTYPE
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MEDIUM ROWID or UROWID column R-TYPE-06

A stored ROWID is a physical row address, and PostgreSQL has no stable one: ctid changes on update and vacuum. Replace the column with a real key reference; whatever wrote and followed these addresses changes with it.

RECON_TEST.LEGACY_REFS.REF_ADDR	ROWID
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LOW Synonym R-OBJ-06

PostgreSQL has no synonyms. Same-database synonyms usually dissolve into search_path; cross-schema renames become views or schema qualification at the call sites.

RECON_TEST.STAFF	for RECON_TEST.EMP
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LOW Optimizer hint R-PERF-01 • extension: pg_hint_plan

PostgreSQL ignores optimizer hints; pg_hint_plan can pin plans but is a last resort. A hinted statement was tuned by hand against Oracle's planner, so budget a fresh look at its plan and indexes on PostgreSQL instead of a plain rewrite.

RECON_TEST.ARCHIVE_NOTES	optimizer hint (first at line 8)
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LOW Parallel degree set on object R-PERF-03

PostgreSQL has no per-object parallel degree; the planner decides within `max_parallel_workers_per_gather`. Workloads that relied on forced parallel scans need a plan review and possibly session-level parallel settings.

RECON_TEST.SALES

DEGREE 4

LOW Old-style outer join (+) R-SRC-05

The (+) operator does not exist in PostgreSQL. Rewrite as ANSI LEFT or RIGHT JOIN; semantics match except for a few multi-condition corner cases worth testing.

RECON_TEST.V_EMP_DEPT

uses (+) outer join

LOW SYSDATE R-SRC-11

SYSDATE is evaluated at call time; `now()` is fixed at transaction start. Map to `statement_timestamp()` or `clock_timestamp()` where the difference matters, for example inside long transactions and loops.

RECON_TEST.ARCHIVE_NOTES

SYSDATE (first at line 19)

RECON_TEST.PKG_LEDGER

SYSDATE (first at line 7)

RECON_TEST.TRG_EMP_SAL_GUARD

SYSDATE (first at line 8)

LOW DBMS_OUTPUT usage R-SYS-05 • extension: oraface

Map `PUT_LINE` to `RAISE NOTICE` for diagnostics. If callers consume the output programmatically, that pattern needs a real interface instead.

RECON_TEST.ARCHIVE_NOTES

DBMS_OUTPUT (first at line 19)

LOW **NUMBER without precision** R-TYPE-04

Bare NUMBER accepts up to 38 significant digits. Map to numeric, and check whether application code assumes integer behavior that a typed bigint would express better.

RECON_TEST.MV_DEPT_SALARIES.HEADCOUNT	NUMBER with no precision
RECON_TEST.MV_DEPT_SALARIES.TOTAL_SAL	NUMBER with no precision
RECON_TEST.V_ORG_CHART.DEPTH	NUMBER with no precision

INFO **Invalid object** R-OBJ-08

Already broken before any migration. Decide per object: fix it, or exclude it from scope in writing so the estimate does not pay for dead code.

RECON_TEST.ARCHIVE_NOTES	status INVALID
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INFO **Swallowed exception** R-SRC-07

WHEN OTHERS THEN NULL hides every failure including the ones a migration introduces. Worth removing before the port so the test phase can actually see errors.

RECON_TEST.ARCHIVE_NOTES	WHEN OTHERS THEN NULL (first at line 22)
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INFO **Table without optimizer statistics** R-STAT-01

num_rows is null, so data-volume estimates for this table are unreliable. Gather statistics before trusting sizing numbers.

RECON_TEST.AUDIT_LOG	no statistics gathered
RECON_TEST.DEPT	no statistics gathered
RECON_TEST.EMP	no statistics gathered
RECON_TEST.INVOICES	no statistics gathered
RECON_TEST.LEGACY_NOTES	no statistics gathered
RECON_TEST.LEGACY_REFS	no statistics gathered
RECON_TEST.MV_DEPT_SALARIES	no statistics gathered
RECON_TEST.PRODUCT_SPECS	no statistics gathered

... and 2 more (10 total; complete list in the findings CSV).

INFO **String lengths declared in bytes** R-TYPE-09

VARCHAR2 and CHAR columns declared without CHAR semantics limit bytes, and in a multibyte database a 30-byte column holds fewer than 30 characters of non-ASCII text. PostgreSQL lengths count characters, so the converted columns accept more text than Oracle did; add CHECK (octet_length(col) <= n) where the byte limit carried meaning, and expect NLS_LENGTH_SEMANTICS to stop mattering.

RECON_TEST.16 string columns	declared in BYTE semantics; PostgreSQL counts characters
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Risk outlook

Of the 50 findings, 8 are classified as high severity, 18 as medium, 11 as low, and 13 as informational. The high-severity items include a database link, package-level state, UTL_FILE usage, a compound trigger, an autonomous transaction, a LONG or LONG RAW column, and a BFILE column, each representing constructs with no direct PostgreSQL equivalent and requiring deliberate redesign. The medium-severity findings, which form the largest group, include scheduler jobs, materialized views, user-defined object types, dynamic SQL, and sequence-assigned keys via triggers, all of which are addressable through established migration patterns but require careful validation. The low and informational findings, including missing optimizer statistics on ten tables and several SYSDATE and NUMBER precision observations, are lower risk but should not be deferred indefinitely, as they can affect performance and data fidelity after cutover.

Recommended approach

Given the severity mix, in which high-severity findings are concentrated in a minority of constructs while the bulk of the work lies in medium-severity items, the recommended approach is to phase the migration by addressing the seven high-severity findings first, as these carry the greatest architectural risk and the least direct portability. Analysis of the hotspot distribution shows that a small number of objects, notably object-1 and object-2, account for a disproportionate share of findings and weighted risk, suggesting that early remediation effort should be concentrated on these objects to reduce overall project risk efficiently. Subsequent phases can address the remaining medium-severity findings in a systematic pass across the codebase, followed by low-severity and informational cleanup, including statistics gathering and datatype precision review, which can be scheduled alongside testing and validation activities without blocking the critical path.

Effort estimate

COMPONENT	PERSON-DAYS
baseline and environment	5.0
schema conversion	1.7
finding remediation	63.2
PL/SQL porting by volume	0.8
data movement	0.0
Development subtotal	70.7

With testing and stabilization: **92 / 113 / 156** person-days (low / expected / high).

Assumptions

- calibration profile 'default': the public engine's conservative defaults, unchanged
- stored code ports at 200 lines per person-day including unit tests
- repeated findings of one rule cost a severity-dependent fraction of the first fix
- testing and stabilization runs 1.3x to 2.2x development, 1.6x expected
- static estimate from schema facts alone: workload, data quality, and application coupling are not visible here

Appendix: findings detail

Each rule lists up to twelve objects here so repeated findings stay readable at estate scale; the findings CSV delivered with this report carries every object with its evidence.

RULE	SEVERITY	OBJECT	EVIDENCE
R-OBJ-01	high	RECON_TEST.LOOPBACK	to //localhost:1521/XE as RECON_TEST
R-PKG-01	high	RECON_TEST.PKG_LEDGER	2 package-level declaration(s), first at line 2
R-PKG-01	high	RECON_TEST.PKG_LEDGER (body)	1 package-level declaration(s), first at line 2
R-SYS-01	high	RECON_TEST.ARCHIVE_NOTES	UTL_FILE (first at line 16)
R-TRG-01	high	RECON_TEST.TRG_EMP_SAL_GUARD	on EMP
R-TRG-02	high	RECON_TEST.TRG_EMP_AUDIT	PRAGMA AUTONOMOUS_TRANSACTION (first at line 5)
R-TYPE-01	high	RECON_TEST.LEGACY_NOTES.BODY	LONG
R-TYPE-07	high	RECON_TEST.LEGACY_REFS.SCAN_DOC	BFILE
R-OBJ-02	medium	RECON_TEST.NIGHTLY_ROLLUP	scheduler job
R-OBJ-03	medium	RECON_TEST.MV_DEPT_SALARIES	materialized view
R-OBJ-05	medium	RECON_TEST.T_MONEY	object type
R-PKG-02	medium	RECON_TEST.PKG_LEDGER	BEGIN blocks exceed subprograms by 1 (possible initialization block)
R-SRC-01	medium	RECON_TEST.PKG_LEDGER	EXECUTE IMMEDIATE (first at line 27)
R-SRC-02	medium	RECON_TEST.FIND_MANAGER	GOTO (first at line 14)
R-SRC-03	medium	RECON_TEST.PKG_LEDGER	BULK COLLECT (first at line 24)
R-SRC-04	medium	RECON_TEST.V_ORG_CHART	CONNECT BY in view definition
R-SRC-14	medium	RECON_TEST.TRG_EMP_AUDIT	COMMIT or ROLLBACK (first at line 15)
R-SRC-16	medium	RECON_TEST.PKG_LEDGER	collection type declaration (first at line 20)
R-SRC-18	medium	RECON_TEST.ARCHIVE_NOTES	empty-string literal (first at line 4)
R-SRC-19	medium	RECON_TEST.ARCHIVE_NOTES	ROWID (first at line 3)
R-SYS-04	medium	RECON_TEST.ARCHIVE_NOTES	DBMS_LOB (first at line 14)
R-TAB-01	medium	RECON_TEST.STAGING_ROWS	GLOBAL TEMPORARY
R-TRG-03	medium	RECON_TEST.TRG_EMP_AUDIT	NEXTVAL in trigger (first at line 14)
R-TRG-03	medium	RECON_TEST.TRG_INVOICES_ID	NEXTVAL in trigger (first at line 6)

R-TYPE-02	medium	RECON_TEST.PRODUCT_SPECS.SPEC	XMLTYPE
R-TYPE-06	medium	RECON_TEST.LEGACY_REFS.REF_ADDR	ROWID
R-OBJ-06	low	RECON_TEST.STAFF	for RECON_TEST.EMP
R-PERF-01	low	RECON_TEST.ARCHIVE_NOTES	optimizer hint (first at line 8)
R-PERF-03	low	RECON_TEST.SALES	DEGREE 4
R-SRC-05	low	RECON_TEST.V_EMP_DEPT	uses (+) outer join
R-SRC-11	low	RECON_TEST.ARCHIVE_NOTES	SYSDATE (first at line 19)
R-SRC-11	low	RECON_TEST.PKG_LEDGER	SYSDATE (first at line 7)
R-SRC-11	low	RECON_TEST.TRG_EMP_SAL_GUARD	SYSDATE (first at line 8)
R-SYS-05	low	RECON_TEST.ARCHIVE_NOTES	DBMS_OUTPUT (first at line 19)
R-TYPE-04	low	RECON_TEST.MV_DEPT_SALARIES.HEADCOUNT	NUMBER with no precision
R-TYPE-04	low	RECON_TEST.MV_DEPT_SALARIES.TOTAL_SAL	NUMBER with no precision
R-TYPE-04	low	RECON_TEST.V_ORG_CHART.DEPTH	NUMBER with no precision
R-OBJ-08	info	RECON_TEST.ARCHIVE_NOTES	status INVALID
R-SRC-07	info	RECON_TEST.ARCHIVE_NOTES	WHEN OTHERS THEN NULL (first at line 22)
R-STAT-01	info	RECON_TEST.AUDIT_LOG	no statistics gathered
R-STAT-01	info	RECON_TEST.DEPT	no statistics gathered
R-STAT-01	info	RECON_TEST.EMP	no statistics gathered
R-STAT-01	info	RECON_TEST.INVOICES	no statistics gathered
R-STAT-01	info	RECON_TEST.LEGACY_NOTES	no statistics gathered
R-STAT-01	info	RECON_TEST.LEGACY_REFS	no statistics gathered
R-STAT-01	info	RECON_TEST.MV_DEPT_SALARIES	no statistics gathered
R-STAT-01	info	RECON_TEST.PRODUCT_SPECS	no statistics gathered
R-STAT-01	info	RECON_TEST.SALES	no statistics gathered
R-STAT-01	info	RECON_TEST.STAGING_ROWS	no statistics gathered
R-TYPE-09	info	RECON_TEST.16 string columns	declared in BYTE semantics; PostgreSQL counts characters

Produced with the pgrecon assessment engine from an offline extraction; no connection was made to the source database.
Findings are deterministic and reproducible from the same inventory. DevCrafter · devcrafterai.com